

TAILSPIN CIVIC SYSTEMS, INC. — SECURITY CONTROLS TRACEABILITY MATRIX (SCTM)

AwardWorks Platform (SaaS) | FedRAMP Moderate Baseline | NIST SP 800-53 Rev. 5 | Fictional organization. Created for portfolio demonstration purposes.

| Control ID | Control Family                            | Control Name                                 | Baseline | Implementation Status | Control Origination              | Implementation Summary                                                                                                                                                                                               | Assessment Method        | Assessment Result    | Related POA&M ID |
|------------|-------------------------------------------|----------------------------------------------|----------|-----------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------|------------------|
| AC-2       | Access Control                            | Account Management                           | Moderate | Partially Implemented | Service Provider Corporate       | Okta/HRIS lifecycle automation and account approvals exist, but four batch jobs share one elevated database account. Process attribution and least privilege are not met for that account; see V-008 / SAR FIND-002. | Examine, Interview, Test | Other Than Satisfied | V-008            |
| AC-2(1)    | Access Control                            | Automated System Account Management          | Moderate | Implemented           | Service Provider Corporate       | Okta Lifecycle Management automates provisioning/deprovisioning; API-driven, no manual account creation in production.                                                                                               | Examine, Test            | Satisfied            |                  |
| AC-6(9)    | Access Control                            | Log Use of Privileged Functions              | Moderate | Partially Implemented | Service Provider System-Specific | Privileged actions logged in CloudTrail and forwarded to SIEM; logging coverage gap identified on legacy reporting subsystem.                                                                                        | Examine, Test            | Other Than Satisfied | V-003            |
| AC-17      | Access Control                            | Remote Access                                | Moderate | Implemented           | Service Provider Corporate       | All remote administrative access requires MFA and traverses a bastion host with session recording; direct SSH from internet is blocked by SCP.                                                                       | Examine, Test            | Satisfied            |                  |
| AU-2       | Audit and Accountability                  | Event Logging                                | Moderate | Implemented           | Service Provider System-Specific | Auditable event list documented and approved; covers authentication, authorization, data access, configuration change, and privilege escalation events.                                                              | Examine, Interview       | Satisfied            |                  |
| AU-6       | Audit and Accountability                  | Audit Record Review, Analysis, and Reporting | Moderate | Partially Implemented | Service Provider System-Specific | SIEM correlation rules deployed; documented weekly review procedure exists but review evidence was inconsistent for a two-month period.                                                                              | Examine, Interview       | Other Than Satisfied | V-005            |
| AU-11      | Audit and Accountability                  | Audit Record Retention                       | Moderate | Implemented           | Service Provider System-Specific | Audit records retained online for 90 days and in S3 Glacier for 12 months, meeting FedRAMP retention requirements.                                                                                                   | Examine                  | Satisfied            |                  |
| CA-7       | Assessment, Authorization, and Monitoring | Continuous Monitoring                        | Moderate | Partially Implemented | Service Provider System-Specific | ConMon program documented with monthly vulnerability scanning and reporting; monthly deliverable submitted late in two of the last six cycles.                                                                       | Examine, Interview       | Other Than Satisfied | V-007            |
| CM-2       | Configuration Management                  | Baseline Configuration                       | Moderate | Implemented           | Service Provider System-Specific | Infrastructure defined as code (Terraform); AMIs hardened to CIS Level 1 benchmark and version-controlled.                                                                                                           | Examine, Test            | Satisfied            |                  |
| CM-6       | Configuration Management                  | Configuration Settings                       | Moderate | Partially Implemented | Service Provider System-Specific | CIS benchmark applied to compute baseline; drift detected on three application servers during assessment sampling.                                                                                                   | Test                     | Other Than Satisfied | V-002            |
| CM-7       | Configuration Management                  | Least Functionality                          | Moderate | Implemented           | Service Provider System-Specific | Container images built from minimal base images; unnecessary services and ports disabled by default and enforced via admission controller.                                                                           | Examine, Test            | Satisfied            |                  |
| CP-9       | Contingency Planning                      | System Backup                                | Moderate | Implemented           | Service Provider System-Specific | Automated encrypted backups: database PITR (5-min RPO) plus daily full snapshots replicated cross-region.                                                                                                            | Examine, Test            | Satisfied            |                  |
| CP-10      | Contingency Planning                      | System Recovery and Reconstitution           | Moderate | Partially Implemented | Service Provider System-Specific | Recovery procedures documented; full-system recovery has not been exercised within the required annual period.                                                                                                       | Examine, Interview       | Other Than Satisfied | V-004            |
| IA-2(1)    | Identification and Authentication         | MFA to Privileged Accounts                   | Moderate | Implemented           | Service Provider Corporate       | Phishing-resistant MFA (FIDO2/WebAuthn) enforced for all privileged accounts via Okta policy.                                                                                                                        | Examine, Test            | Satisfied            |                  |
| IA-5       | Identification and Authentication         | Authenticator Management                     | Moderate | Implemented           | Service Provider Corporate       | Password policy aligned to NIST SP 800-63B; no forced periodic rotation, breached-password screening enabled.                                                                                                        | Examine, Test            | Satisfied            |                  |
| IR-4       | Incident Response                         | Incident Handling                            | Moderate | Implemented           | Service Provider Corporate       | Documented IR plan with defined severity tiers; 24x7 on-call rotation; US-CERT reporting procedure defined per FedRAMP requirements.                                                                                 | Examine, Interview       | Satisfied            |                  |
| IR-6       | Incident Response                         | Incident Reporting                           | Moderate | Partially Implemented | Service Provider Corporate       | Internal reporting path defined; the one-hour US-CERT/FedRAMP ISSO notification requirement is documented but has not been exercised in a tabletop.                                                                  | Examine, Interview       | Other Than Satisfied | V-006            |
| RA-5       | Risk Assessment                           | Vulnerability Monitoring and Scanning        | Moderate | Partially Implemented | Service Provider System-Specific | Authenticated scanning performed monthly on infrastructure, OS, web, and database layers; several High findings exceeded the 30-day FedRAMP remediation window.                                                      | Examine, Test            | Other Than Satisfied | V-001            |
| SA-11      | System and Services Acquisition           | Developer Testing and Evaluation             | Moderate | Implemented           | Service Provider Corporate       | SAST and dependency scanning enforced in CI; annual third-party penetration test performed with findings tracked to closure.                                                                                         | Examine                  | Satisfied            |                  |

| Control ID | Control Family                        | Control Name                               | Baseline | Implementation Status | Control Origination                     | Implementation Summary                                                                                                                                     | Assessment Method  | Assessment Result    | Related POA&M ID |
|------------|---------------------------------------|--------------------------------------------|----------|-----------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------|------------------|
| SC-7       | System and Communications Protection  | Boundary Protection                        | Moderate | Implemented           | Service Provider System-Specific        | Authorization boundary enforced via VPC segmentation, security groups, WAF, and a deny-by-default egress policy; managed interfaces documented in the SSP. | Examine, Test      | Satisfied            |                  |
| SC-8       | System and Communications Protection  | Transmission Confidentiality and Integrity | Moderate | Implemented           | Service Provider System-Specific        | TLS 1.2+ with FIPS 140-3 validated cryptographic modules enforced on all external and internal service-to-service communication.                           | Examine, Test      | Satisfied            |                  |
| SC-13      | System and Communications Protection  | Cryptographic Protection                   | Moderate | Implemented           | Service Provider System-Specific        | FIPS 140-3 validated modules used for all cryptographic operations; AWS KMS FIPS endpoints in use.                                                         | Examine            | Satisfied            |                  |
| SC-28      | System and Communications Protection  | Protection of Information at Rest          | Moderate | Implemented           | Service Provider System-Specific        | All data stores encrypted at rest with AES-256 via customer-managed KMS keys with annual rotation.                                                         | Examine, Test      | Satisfied            |                  |
| SI-2       | System and Information Integrity      | Flaw Remediation                           | Moderate | Partially Implemented | Service Provider System-Specific        | Patching automated for OS/container layers; remediation SLA missed for a subset of High-severity findings (see RA-5).                                      | Examine, Test      | Other Than Satisfied | V-001            |
| SI-4       | System and Information Integrity      | System Monitoring                          | Moderate | Implemented           | Service Provider System-Specific        | IDS/IPS via GuardDuty, host-based monitoring, and file integrity monitoring feeding the SIEM with 24x7 alerting.                                           | Examine, Test      | Satisfied            |                  |
| PS-3       | Personnel Security                    | Personnel Screening                        | Moderate | Implemented           | Service Provider Corporate              | Background investigations completed for all personnel with system access prior to provisioning; re-screening every five years.                             | Examine, Interview | Satisfied            |                  |
| AT-2       | Awareness and Training                | Literacy Training and Awareness            | Moderate | Implemented           | Service Provider Corporate              | Annual security awareness training with role-based modules for privileged users; completion tracked at 100% for in-scope personnel.                        | Examine            | Satisfied            |                  |
| MP-6       | Media Protection                      | Media Sanitization                         | Moderate | Inherited             | Inherited from AWS (FedRAMP High P-ATO) | Media sanitization is inherited from the underlying IaaS provider; customer responsibility limited to logical data deletion.                               | Examine            | Satisfied            |                  |
| PE-3       | Physical and Environmental Protection | Physical Access Control                    | Moderate | Inherited             | Inherited from AWS (FedRAMP High P-ATO) | Physical access to data centers is fully inherited from the underlying IaaS provider's authorized environment.                                             | Examine            | Satisfied            |                  |
| PL-2       | Planning                              | System Security and Privacy Plans          | Moderate | Implemented           | Service Provider System-Specific        | This SSP documents the authorization boundary, control implementations, and system environment; reviewed and updated at least annually.                    | Examine            | Satisfied            |                  |

# CONTROL IMPLEMENTATION SUMMARY

Counts derived from the SCTM tab. Full FedRAMP Moderate baseline is 32

| Implementation Status | Count | % of Sampled Controls |
|-----------------------|-------|-----------------------|
| Implemented           | 21    | 63.6%                 |
| Partially Implemented | 9     | 27.3%                 |
| Inherited             | 2     | 6.1%                  |
| Planned               | 0     | 0.0%                  |
| Not Applicable        | 1     | 3.0%                  |
| TOTAL                 | 33    |                       |

## Assessment Result

| Result               | Count | % of Sampled Controls |
|----------------------|-------|-----------------------|
| Satisfied            | 24    | 72.7%                 |
| Other Than Satisfied | 9     | 27.3%                 |

# SCTM — LEGEND & METHODOLOGY

|                       |                                                                                                                                                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose               | The Security Controls Traceability Matrix documents how each NIST SP 800-53 Rev. 5 control in the FedRAMP Moderate baseline is implemented, who is responsible for it, how it was assessed, and whether it passed.                                                                                             |
| Implementation Status | Implemented   Partially Implemented   Planned   Alternative Implementation   Not Applicable   Inherited (from an underlying authorized service provider)                                                                                                                                                       |
| Control Origination   | Service Provider Corporate (org-wide control)   Service Provider System-Specific (unique to this system)   Inherited   Customer (Configured/Provided)   Shared                                                                                                                                                 |
| Assessment Methods    | Examine (documentation/artifacts), Interview (personnel), Test (technical validation) — per NIST SP 800-53A                                                                                                                                                                                                    |
| Assessment Result     | Satisfied = control operating as designed   Other Than Satisfied = deficiency identified, which generates a POA&M item<br>Every 'Other Than Satisfied' control links to a POA&M ID in the Related POA&M ID column, so a reviewer can trace a control deficiency straight to its remediation plan and due date. |
| Traceability          |                                                                                                                                                                                                                                                                                                                |
| Scope note            | This sample documents 33 representative controls across 16 families, including one Not Applicable determination (SC-15). A production FedRAMP Moderate SSP addresses all 323 controls and enhancements.                                                                                                        |